

Students Name:	Grade: MYP1	Date:
Subject: Mathematics	Unit Title: 1. Numerical Reasoning 2. Reasoning with Data	
Task: Term 1_Sample Paper		

1. The school recorded the distances completed by students during a fun run. The results were recorded in different forms to help students analyse and compare the data.

Data Set: 0, 3, 5.7, 8, 12.4, 15, -2, 6.25, 20, -5.5, 10

a) **Identify** the numbers from the data set that are:

- i. Natural numbers
- ii. Whole numbers
- iii. Integers
- iv. Decimals

b) **Recognise** the numbers that belong to both the whole-number and integer sets.

c) **Order** the numbers from least to greatest.

d) **Apply** your understanding of place value to round 18.76 to the nearest tenth.

2. **Recognise** the missing number and the property used.

a) $36 + 14 = 14 + \underline{\hspace{2cm}}$ b) $8 \times 12 = \underline{\hspace{2cm}} \times 8$ c) $(5 + 7) + 3 = 5 + (7 + \underline{\hspace{2cm}})$ d) $9 \times (20 + 4) = (9 \times 20) + (9 \times \underline{\hspace{2cm}})$ e) $45 + \underline{\hspace{2cm}} = 45$ f) $32 \times \underline{\hspace{2cm}} = 32$ g) $56 \times \underline{\hspace{2cm}} = 0$	
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3. During a mountain expedition, students recorded changes in altitude. Positive integers represent movement upwards and negative integers represent movement downwards.

Altitude changes recorded: +8 m, -5 m, +12 m, -7 m, +4 m

a. Identify the integer that represents a descent of 7 metres. b. Determine the final altitude change after the group moves: +15 m, -8 m, +6 m, -10 m. c. Apply the BODMAS rule to evaluate the complete expression. Show all steps clearly. $20 - 2^3 \times (-3) + 18 \div 3$	b. Calculate the following: i. $+8 + (-5) + 8 + (-5)$ ii. $-7 + (+12) - 7 + (+12)$ iii. $-6 - (+4) - 6 - (+4)$ iv. $9 - (-5) + 11 - 2$
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4. A school is preparing activity stations. Three activities repeat at different intervals:

- Math Puzzle Station: every 12 minutes
- Logic Game Station: every 18 minutes
- Number Challenge Station: every 24 minutes

At 9:00 a.m., all three activities begin together.

a) Determine the prime factorisation of 12, 18 and 24. b) Calculate the LCM of 12, 18 and 24.	
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c) Apply your answer to determine the next time when all three activities will begin together. d) Explain why the LCM is used to solve this problem	
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5. The school has 48 red cards and 72 blue cards. The teacher wants to make identical sets using all the cards.

a) Calculate the HCF of 48 and 72. b) Determine the greatest number of identical sets that can be made. c) Calculate the number of red and blue cards in each set. d) Justify your answer by showing how the HCF helps in making the largest possible number of equal sets.	
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6. The school is designing a square flower bed and a cube-shaped storage box for the garden. Students need to use their understanding of squares, square roots, cubes and cube roots to determine the dimensions.

a) A square flower bed has an area of 196 m^2. Calculate the length of one side of the flower bed. b) A cube-shaped storage box has a volume of 343 cm^3. Determine the length of one edge of the box. c) Explain the relationship between cubing a number and finding its cube root using an example.	
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7. The Mathematics Club is organising a charity fair. Students buy and sell handmade products and offer discounts to attract customers. They need to calculate profit, loss, profit percentage, loss percentage and discount.

a. Identify the cost price, selling price, profit/loss, and discount in the following situation: A bag has a cost price of ₹800 and is sold for ₹680 after a discount. b. Calculate the discount amount and discount percentage on the bag. c. Determine the profit or loss when an item costing ₹500 is sold for ₹575. Also calculate the profit percentage.	
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- d. **Calculate** the selling price of a jacket marked at ₹1,200 after a 15% discount.

8. A shop offers a 20% discount on all the following items:

Item	Cost Price	Discount	Discount Amount	Selling Price
Notebook set	₹250	20%		
School bag	₹500	20%		
Water bottle	₹750	20%		
Art kit	₹1000	20%		

- a) **Organise** the information by calculating the discount amount and selling price for each item.
- b) **Describe** the pattern you notice between the cost price, discount amount and selling price.
- c) **Generalise** a rule for finding the selling price of an item after a 20% discount.
- d) **Verify** your rule using an item with a cost price of ₹1,500.

9. Students are preparing fruit punch for a school event. The recipe uses orange juice and pineapple juice in the ratio 3 : 2.

a) **Organise** the following quantities in a table and complete the missing values.

Orange Juice (L)	Pineapple Juice (L)
3	2
6	
	6
12	
	10

Rough work

- b) **Describe** the pattern you notice between the quantities of orange juice and pineapple juice.
- c) **Generalise** a rule that can be used to find the amount of pineapple juice needed for any quantity of orange juice.
- d) **Apply** your rule to determine the amount of each juice needed to make 25 L of fruit punch while maintaining the same ratio.

10. a) Identify whether each of the following is written in 12-hour or 24-hour format:

- i) 7:25 a.m.
- ii) 18:40
- ii) 11:15 p.m.
- iv) 06:50

b) **Determine** the time 2 hours 35 minutes after 16:20.

c) **Explain** why 12:30 p.m. is written as 12:30 in the 24-hour format, but 1:30 p.m. is written as 13:30.

11. The school librarian surveyed 20 MYP 1 students to understand their reading habits. Each student recorded the number of minutes they read in one day.

Data Set : 15, 20, 25, 20, 30, 15, 35, 25, 20, 40,
30, 25, 15, 20, 35, 30, 25, 20, 30, 25

Reading Time (Minutes)	Tally Marks	Frequency
15		
20		
25		
30		
35		
40		
Total		

a) **i) Represent** the reading-time data using a bar graph. Include:

- an appropriate title
- labelled axes
- a suitable scale
- accurately plotted bars

OR

ii) Construct a line plot to represent the reading-time data. Use an appropriate scale and mark each observation clearly. **Interpret** and **state** the reading time that occurs most frequently.

b) **Calculate** the mean, median, mode and range of the reading times.

c) **Analyse** the mean, median and mode and **explain** which measure best represents the typical reading time. Give a reason based on the data.

12. The same 20 students were asked about their favourite free-time activity.

Activity	Number of Students
Sports	6
Reading	5
Music	4
Art	3
Gaming	2
Total	20

- a) **Calculate** the angle for each category.
- b) **Represent** the data using a pie chart. Clearly label each sector with the activity and its corresponding value.
- c) **Analyse** both data sets and **explain** one conclusion the school could make about students' reading habits and free-time preferences.

13. A weather station recorded the weather conditions for 20 days during a month.

Weather Condition	Number of days
Sunny	8
Cloud	5
Rainy	4
Windy	2
Stormy	1
Total	20

a) **Calculate** the probability of each in its simplest fraction form.

- a sunny day
- a cloudy day
- a rainy day
- a windy day
- a stormy day

b) State which weather condition has the greatest probability and which has the smallest probability.	
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